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import pandas as pd

# Load dataset
df = pd.read_csv("upi_transactions_2024.csv")

print("=" * 60)
print("STAGE 5: DATA AGGREGATION & REPRESENTATION")
print("=" * 60)

# 1. Dataset Information
print("\n1. DATASET")
print("Rows      :", df.shape[0])
print("Columns   :", df.shape[1])

# 2. Transaction Type Aggregation
transaction_type_summary = (
    df.groupby("transaction type")
      .agg(
          transaction_count=("transaction id", "count"),
          total_amount=("amount (INR)", "sum"),
          average_amount=("amount (INR)", "mean")
      )
      .reset_index()
)

print("\n2. TRANSACTION TYPE SUMMARY")

for _, row in transaction_type_summary.iterrows():
    print("\nTransaction Type :", row["transaction type"])
    print("Transaction Count:", int(row["transaction_count"]))
    print("Total Amount      :", round(row["total_amount"], 2))
    print("Average Amount    :", round(row["average_amount"], 2))

# 3. Merchant Category Aggregation
merchant_summary = (
    df.groupby("merchant_category")
      .agg(
          transaction_count=("transaction id", "count"),
          total_amount=("amount (INR)", "sum"),
          average_amount=("amount (INR)", "mean")
      )
      .reset_index()
      .sort_values("transaction_count", ascending=False)
)

print("\n3. MERCHANT CATEGORY SUMMARY")

for _, row in merchant_summary.iterrows():
    print("\nMerchant Category :", row["merchant_category"])
    print("Transaction Count :", int(row["transaction_count"]))
    print("Total Amount      :", round(row["total_amount"], 2))
    print("Average Amount    :", round(row["average_amount"], 2))

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STAGE 5: DATA AGGREGATION & REPRESENTATION
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1. DATASET

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Rows      : 250000
Columns   : 17

```

2. TRANSACTION TYPE SUMMARY

```

Transaction Type : Bill Payment
Transaction Count: 37368
Total Amount     : 48895743
Average Amount   : 1308.49

```

```

Transaction Type : P2M
Transaction Count: 87660
Total Amount     : 115717567
Average Amount   : 1320.07

```

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Transaction Type : P2P
Transaction Count: 112445
Total Amount     : 147154648
Average Amount   : 1308.68

```

Transaction Type : Recharge
Transaction Count: 12527
Total Amount : 16171051
Average Amount : 1290.9

3. MERCHANT CATEGORY SUMMARY

Merchant Category : Grocery
Transaction Count : 49966
Total Amount : 58277893
Average Amount : 1166.35

Merchant Category : Food
Transaction Count : 37464
Total Amount : 19919402
Average Amount : 531.69

Merchant Category : Shopping
Transaction Count : 29872
Total Amount : 76863207
Average Amount : 2573.09

Merchant Category : Fuel
Transaction Count : 25063
Total Amount : 38982575
Average Amount : 1555.38

Merchant Category : Other
Transaction Count : 24828
Total Amount : 21073449
Average Amount : 848.78

Merchant Category : Utilities

```
# 4. Fraud Aggregation
fraud_summary = (
    df.groupby("fraud_flag")
        .agg(
            transaction_count=("transaction id", "count"),
            total_amount=("amount (INR)", "sum"),
            average_amount=("amount (INR)", "mean")
        )
        .reset_index()
)

print("\n4. FRAUD FLAG SUMMARY")

for _, row in fraud_summary.iterrows():
    status = "Fraudulent" if row["fraud_flag"] == 1 else "Non-Fraudulent"

    print("\nStatus          :", status)
    print("Transaction Count:", int(row["transaction_count"]))
    print("Total Amount      :", round(row["total_amount"], 2))
    print("Average Amount    :", round(row["average_amount"], 2))

# 5. State Aggregation
state_summary = (
    df.groupby("sender_state")
        .agg(
            transaction_count=("transaction id", "count"),
            total_amount=("amount (INR)", "sum")
        )
        .reset_index()
        .sort_values("transaction_count", ascending=False)
)

print("\n5. TOP 5 SENDER STATES")

for _, row in state_summary.head(5).iterrows():
    print("\nState          :", row["sender_state"])
    print("Transaction Count:", int(row["transaction_count"]))
    print("Total Amount      :", round(row["total_amount"], 2))
```

4. FRAUD FLAG SUMMARY

Status : Non-Fraudulent
Transaction Count: 249520
Total Amount : 327219378.0
Average Amount : 1311.4

```
Status          : Fraudulent
Transaction Count: 480
Total Amount     : 719631.0
Average Amount   : 1499.23
```

5. TOP 5 SENDER STATES

```
State           : Maharashtra
Transaction Count: 37427
Total Amount     : 49043948
```

```
State           : Uttar Pradesh
Transaction Count: 30125
Total Amount     : 40035717
```

```
State           : Karnataka
Transaction Count: 29756
Total Amount     : 38451158
```

```
State           : Tamil Nadu
Transaction Count: 25367
Total Amount     : 33343518
```

```
State           : Delhi
Transaction Count: 24870
Total Amount     : 32689865
```

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# 6. Hourly Aggregation
hourly_summary = (
    df.groupby("hour_of_day")
      .agg(
          transaction_count=("transaction id", "count"),
          total_amount=("amount (INR)", "sum")
      )
      .reset_index()
      .sort_values("transaction_count", ascending=False)
)

print("\n6. TOP 5 TRANSACTION HOURS")

for _, row in hourly_summary.head(5).iterrows():
    print("\nHour          :", int(row["hour_of_day"]))
    print("Transaction Count:", int(row["transaction_count"]))
    print("Total Amount      :", round(row["total_amount"], 2))

# 7. Representation Summary
print("\n" + "=" * 60)
print("7. AGGREGATED DATASETS CREATED")
print("=" * 60)

print("Transaction Type Summary :", transaction_type_summary.shape)
print("Merchant Category Summary:", merchant_summary.shape)
print("Fraud Summary           :", fraud_summary.shape)
print("State Summary           :", state_summary.shape)
print("Hourly Summary           :", hourly_summary.shape)

# Completion
print("\n" + "=" * 60)
print("STAGE 5 COMPLETED")
print("=" * 60)
```

6. TOP 5 TRANSACTION HOURS

```
Hour           : 19
Transaction Count: 21232
Total Amount     : 28223522
```

```
Hour           : 18
Transaction Count: 20064
Total Amount     : 26297174
```

```
Hour           : 20
Transaction Count: 18506
Total Amount     : 23822014
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```
Hour           : 17
Transaction Count: 18340
Total Amount     : 24514186
```

Hour : 12
Transaction Count: 17516
Total Amount : 23406280

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7. AGGREGATED DATASETS CREATED
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Transaction Type Summary : (4, 4)
Merchant Category Summary: (10, 4)
Fraud Summary : (2, 4)
State Summary : (10, 3)
Hourly Summary : (24, 3)

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STAGE 5 COMPLETED
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